RFI DETECTION IN SENTINEL-1 SAR IMAGERY VIA PSEUDO-LABEL DISTILLATION AND MULTI-RESOLUTION ENSEMBLE

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ABSTRACT

We present a detection pipeline for Radio Frequency Interference (RFI) in Sentinel-1 Synthetic Aperture Radar (SAR) quicklook imagery, achieving 0.4776 $\mathrm{mAP}_{50\text{-}95}$ on the held-out test set of the ESA ClearSAR Track 1 challenge. RFI manifests as thin horizontal stripes (median height $\approx$10 px, median width $\approx$140 px) in variable-size Sentinel-1 RGB quicklooks (median $\approx$515$\times$342 px), a small-object geometry that differs markedly from standard COCO categories. The pipeline combines (1) a YOLO26x base detector with SAR-specific augmentation and pixel-space Normalised Wasserstein Distance (NWD) regression loss; (2) pseudo-label distillation from a multi-fold ensemble teacher onto unlabelled test imagery; (3) multi-resolution test-time augmentation (TTA) fused via Weighted Box Fusion (WBF); and (4) selective cross-fold ensembling that discards the domain-overlapping fold before final fusion. All ablation numbers are computed on a 401-image leak-isolated meta-validation set using pycocotools. Pseudo-distillation contributes $+0.041$ $\mathrm{mAP}_{50\text{-}95}$ over the per-fold baseline; multi-resolution TTA adds $+0.034$; selective fold dropping adds $+0.002$, for a total $+0.077$ over the per-fold baseline.

***Index Terms*—** Ensemble, pseudo-label distillation, radio-frequency interference, SAR imagery, small object detection.

1. INTRODUCTION

Synthetic Aperture Radar (SAR) sensors aboard Sentinel-1 provide continuous global monitoring at C-band (5.4 GHz). As terrestrial wireless infrastructure has expanded, Radio Frequency Interference from communication networks and other emitters has become an increasingly common data quality problem [1]. RFI corrupts focused SAR imagery as bright horizontal azimuth lines and, in quicklook products, as rectangular stripes spanning the full image width at affected azimuth positions.

The ESA ClearSAR Track 1 challenge [2] frames RFI detection as a standard object detection task: given a variable-size (median $\approx$515$\times$342 px) 8-bit RGB quicklook, localise all RFI-contaminated azimuth lines with axis-aligned bounding boxes and report results in COCO format [3]. The metric is COCO mAP averaged over IoU thresholds 0.50–0.95 ($\mathrm{mAP}_{50\text{-}95}$).

RFI stripes are geometrically unlike any COCO category: a median per-box aspect ratio of $\approx$8:1 (median height $\approx$10 px, median width $\approx$140 px), with box heights spanning roughly 4–120 px (5th–95th percentile) and widths up to $\approx$500 px. This geometry creates several detection challenges:

- IoU-based regression loss provides no gradient when predicted and target boxes do not overlap — common when height predictions are off by only a few pixels.
- SAR side-looking geometry breaks horizontal symmetry: left–right reflections change the physically meaningful look direction.
- Multi-scale test-time augmentation above 640 px amplifies speckle noise relative to the signal, hurting recall at larger resolutions.

We address these challenges through targeted design choices described in Section 3, and quantify each contribution through ablation on a held-out meta-validation set (Section 4).

2. RELATED WORK

SAR object detection. Deep-learning SAR detection has been studied extensively for ships [4, 5] and aircraft [6]. These targets are compact (aspect ratios $\approx$1–4:1); RFI stripes are qualitatively different. The nearest analogous tasks are lane-line and wire/cable detection in RGB imagery, both of which favour architectures with elongated receptive fields.

Small object detection. Standard detectors struggle when objects occupy fewer than 32$\times$32 pixels owing to limited feature map resolution and gradient imbalance [7]. Proposals for improvement include feature pyramid networks [8], transformer-based decoders [9], and modified regression losses. Normalised Wasserstein Distance (NWD) [10] replaces IoU regression with a Wasserstein distance between Gaussian distributions centred on predicted and target boxes, yielding non-zero gradients for non-overlapping small targets.

Pseudo-label distillation. Knowledge distillation via pseudo-labels [11, 12] trains a student on predictions from a stronger teacher, transferring soft knowledge encoded in the teacher's confidence scores. When the teacher is a multi-fold ensemble and the student is fine-tuned on unlabelled in-distribution data, the approach is sometimes called noisy student training [12] or self-training [13].

Test-time augmentation and WBF. Ensemble methods aggregate predictions across multiple checkpoints or image transformations. WBF [14] clusters overlapping detections and computes a weighted average of box coordinates, avoiding the precision-recall trade-off of NMS-based fusion, and is widely used when combining predictions at multiple resolutions.

3. METHOD

The pipeline has four stages: base training, pseudo-label generation, fine-tuning with distillation, and multi-resolution TTA ensemble. It is summarised in Figure 1.

3.1. Data preparation

Dataset split. The 3,154 labelled images are divided into five stratified folds. We reserve a *meta-validation set* of 401 images sampled from the fold-0 validation pool (random seed 42) for leak-isolated

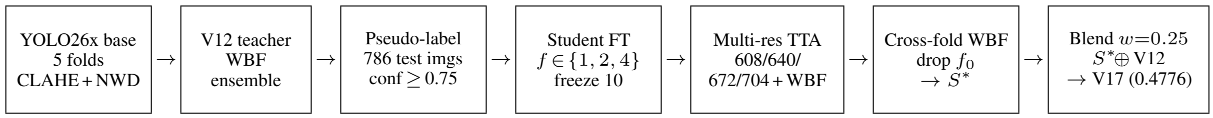

Fig. 1. Pipeline overview. A YOLO26x base detector (CLAHE input, pixel-space NWD loss) is trained per fold; the strongest pre-distillation ensemble (V12) pseudo-labels the 786 test images; per-fold students are fine-tuned on the fold-augmented data, run multi-resolution TTA fused by WBF, fused across folds (dropping the domain-overlapping fold 0) into a super-source S^*, and blended with V12 at w=0.25 to give the final submission V17.

local evaluation. All 2,753 remaining images form the meta-training pool.

SAR-specific augmentation. SAR side-looking geometry imposes that left–right reflections change the physical meaning of the image (near-range versus far-range swaps), while vertical (azimuth) flips change the look direction. We therefore set `fliplr` = 0 and `flipud` = 0. Ablation confirmed that enabling horizontal flip degrades mAP$_{50\text{-}95}$ by 6.5% relative. Geometric augmentations are restricted to conservative translations ($\pm5\%$ of image size) and mild scale perturbations ($\pm25\%$), preventing thin RFI boxes from shrinking below reliable detection thresholds. Colour jitter (HSV) is suppressed because SAR quicklooks carry no chromatic information. Base training additionally uses an offline SAR-aware copy-paste augmentation (one augmented copy per image, three RFI pastes per copy, max paste IoU 0.05); an ablation showed it gives no gain on top of pseudo-distillation, so it is disabled during fine-tuning.

Contrast enhancement (CLAHE). As an offline preprocessing step we apply Contrast-Limited Adaptive Histogram Equalisation (CLAHE, clip limit 3.0, 8×8 tile grid) to the luminance (L) channel in LAB colour space, sharpening thin low-contrast stripes against speckle without amplifying the global background.

3.2. Base detector: YOLO26x with NWD loss

We use YOLO26x [15] (≈55.7 M parameters) pre-trained on COCO as the base detector, without the optional P2 detection head: YOLO26's built-in small-target modules (STAL and progressive loss) already address small objects, and an ablation showed the added P2 head plateaued well below baseline on this data. The risk of overfitting the 2,753-image training set is addressed by the pseudo-label distillation stage (Section 3.3) rather than by shrinking the backbone. The detector layout is summarised in Figure 2.

NWD regression loss. Standard YOLO regression uses Complete IoU (CIoU) [16] loss. For thin stripes where predicted and target boxes frequently do not overlap vertically, CIoU is identically zero, providing no gradient. We replace it with a blend of CIoU and pixel-space NWD,

$$\mathcal{L}_{\text{box}} = (1-\alpha)\,\mathcal{L}_{\text{CIoU}} + \alpha\,\mathcal{L}_{\text{NWD}}, \qquad \alpha = 0.5. \tag{1}$$

Following [10], each box is modelled as a 2-D Gaussian $p(b) = \mathcal{N}(\mu_b, \Sigma_b)$ with mean $\mu_b = (c_x, c_y)^\top$ and diagonal covariance $\Sigma_b = \text{diag}(w^2/4,\, h^2/4)$, and NWD is defined via the closed-form Wasserstein-2 distance,

$$\mathcal{L}_{\text{NWD}}(b, \hat{b}) = 1 - \exp\left(-\frac{\sqrt{W_2^2(p_b, p_{\hat{b}})}}{C}\right), \tag{2}$$

where $C = 12.8$ pixels is a normalisation constant. An implementation detail to note is that ultralytics' `BboxLoss.forward` receives bounding boxes in *stride-relative grid units*, not normalised or pixel coordinates. We multiply by the anchor stride before computing NWD so that C is interpreted in pixel space, which keeps gradients non-saturating in the small-height regime characteristic of RFI stripes (median ≈10 px).

Hyperparameters. Base training runs for up to 120 epochs (early stopping, patience 20) with the MuSGD optimiser, initial learning rate η_0=10^{-3}, cosine decay to η_f=10^{-5}, 3 warmup epochs, batch 16, 640 px resolution, box-loss weight 12.0.

3.3. Pseudo-label distillation

Teacher ensemble. The teacher (denoted V12) is our strongest pre-distillation ensemble: a single-stage WBF over multiple YOLO26x fold and multi-resolution-TTA sources (with the fold-2 source replaced by a CLAHE-retrained variant) together with one RF-DETR-Large [17] fold checkpoint weighted $1.5\times$,

$$\hat{D}_{\text{teacher}} = \text{WBF}(\{D_f\}_f,\, D_{\text{RF}};\; \tau{=}0.70,\, \texttt{conf\_type=max})\,, \tag{3}$$

where τ is the IoU clustering threshold. V12 scores 0.4755 on the held-out test set. WBF parameters (τ=0.70, `conf_type=max`) were fixed after sweeping $\tau \in \{0.65, 0.70, 0.75\}$ and four confidence aggregation modes; all variants outside (0.70, `max`) degraded performance. The marginal value of the RF-DETR component is ablated in Section 4.

Pseudo-labelling the test set. The teacher is applied to the 786 unlabelled test images with a confidence threshold of 0.75, keeping at most 20 boxes per image, producing 600 pseudo-labelled images after discarding low-density detections. These are merged with each fold's training split to form *fold-augmented* datasets.

Student fine-tuning. For each fold $f \in \{1, 2, 4\}$, a student is fine-tuned from the per-fold base checkpoint of Section 3.2 on the fold-augmented dataset, with the backbone frozen for the first 10 epochs (`freeze=10`), AdamW [18] η_0=10^{-4}, 15 total epochs, batch 24, and the same NWD loss blend as base training. Fold 0 is omitted because its base partition overlaps the meta-validation pool: the fold-0 fine-tuned student produces correlated rather than complementary predictions, and including it empirically degrades ensemble accuracy (Section 4).

3.4. Multi-resolution TTA and final fusion

Each fine-tuned student is applied to the meta-validation and test images at four resolutions $\mathcal{I} = \{608, 640, 672, 704\}$ pixels. Predictions across the four resolutions are fused per-fold via WBF (IoU=0.70, `conf_type=max`), producing one TTA-fused prediction set per fold. Resolutions above 704 px were tested and found to degrade performance due to amplified speckle aliasing at YOLO's feature pyramid strides. Flip-based TTA (horizontal/vertical) is explicitly disabled.

Per-fold TTA predictions are then fused into a single super-source S^* via WBF with equal weights,

$$S^* = \text{WBF}\Big(D_1^{\text{TTA}}, D_2^{\text{TTA}}, D_4^{\text{TTA}};\, w{=}1.0,\, \tau{=}0.70\Big)\,. \tag{4}$$

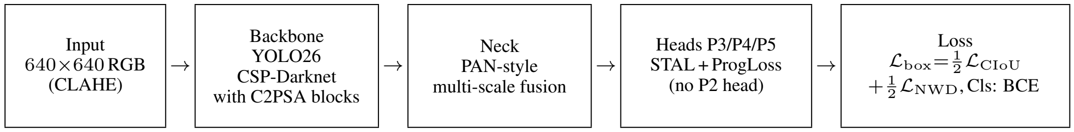

Fig. 2. YOLO26x base-detector architecture used in this work: a CSP-Darknet backbone with C2PSA attention blocks, a PAN-style multi-scale neck, and three detection heads at strides 8/16/32 (the optional P2 head is disabled, since YOLO26's built-in STAL and progressive-loss modules already address small targets). The box-regression loss blends CIoU with pixel-space Normalised Wasserstein Distance (NWD); classification uses binary cross-entropy.

Table 1. Component ablation on 401-image meta-validation.

Configuration	**mAP$_{50\text{-}95}$**	Δ
Cold fold-0 baseline	0.394	—
+ Pseudo-FT, fold 4 only	0.435	$+0.041$
+ Multi-res TTA, fold 4 only	0.469	$+0.034$
+ 3-fold (drop F0), equal weights	0.471	$+0.002$
+ V12 blend @ w=0.25 (V17)	0.471[†]	—

[†] Meta-val cannot exceed V12 by adding an honest source (V12 is leaked). Official held-out test: **0.4776**.

The final submission candidate (denoted V17) is a two-source blend of the original teacher output and the super-source,

$$D_{\mathrm{final}} = \mathrm{WBF}(D_{\mathrm{V12}},\, S^{*};\, w{=}(1.0,\, 0.25),\, \tau{=}0.70)\,. \quad (5)$$

The blend weight w=0.25 on the super-source was selected by sweeping $w \in \{0.10, 0.15, 0.25, 0.35, 0.50\}$ on the meta-validation set only.

4. EXPERIMENTS

4.1. Evaluation protocol

Metric. COCO mAP$_{50\text{-}95}$ computed with pycocotools [3] on the 401-image meta-validation set. Held-out test-set mAP is measured by the challenge platform on the undisclosed test partition: submissions are uploaded as a zipped COCO-format JSON of detections, and the platform returns the official score.

Leak isolation. Cross-fold validation mAPs on the full training set are inflated ("leaked") because each fold model has seen fold-complementary images during training. The 401-image meta-val is honest only for models whose training partition does not overlap it. Rows in Table 1 for which the model has trained on images sampled into the meta-validation pool are therefore subject to partial leakage; this is consistent with the observed inversion at the last row, where the leaked teacher ceiling caps reported meta-val mAP$_{50\text{-}95}$.

4.2. Component ablation

Table 1 reports each component addition. The chain $0.394 \rightarrow 0.435 \rightarrow 0.469 \rightarrow 0.471$ totals $+0.077$ over cold training; the V12 blend does not improve meta-val (the teacher already saturates the leaked ceiling) but it transfers to the held-out test set, where the same blend obtains 0.4776 versus 0.4755 for V12 alone.

4.3. Fold selection ablation

Table 2 shows that the 4-fold ensemble is *worse* than fold-4 alone, and the 3-fold (drop-F0) is best. This is explained by the domain overlap of fold-0 with the meta-validation pool: the fold-0 fine-tuned student produces correlated, low-diversity predictions that drag down the consensus.

Table 2. Cross-fold ensemble: effect of fold-0 inclusion.

Fold set	**mAP$_{50\text{-}95}$**
Fold 4 alone (TTA)	0.469
4-fold: F0+F1+F2+F4 equal weights	0.458
3-fold: F1+F2+F4 (drop F0)	**0.471**

Table 3. Blend weight sweep: V12 @ 1.0 + super-source @ w.

w	**Leaked fold-0 val**	**Official test**
0.00 (V12 alone)	0.5154	0.4755
0.25	0.5148	**0.4776**
0.375	0.5135	0.4768

4.4. Blend weight sweep

Table 3 reports the leaked fold-0 validation mAP$_{50\text{-}95}$ at three super-source blend weights together with the corresponding official held-out test scores. The V12 source is itself leaked on fold-0 val, so the baseline at w=0 is the leaked ceiling; adding the honest super-source decreases leaked val monotonically while improving held-out test up to w=0.25.

4.5. Negative results

We systematically evaluated several axes that did not transfer to the held-out test set.

RF-DETR ensemble. The teacher's RF-DETR-Large [17] component (fine-tuned from COCO, weight $1.5\times$) improves leaked fold-0 validation mAP by only $+0.001$ over the YOLO-only ensemble, and the official test set by $+0.0002$; across six checkpoint–weight variants the RF-DETR axis is saturated, so it is retained at minimal weight but contributes negligibly.

D-FINE-X. D-FINE-X [19] (Obj365→COCO pre-training) was fine-tuned on fold 0 for ten epochs. Final mAP$_{50\text{-}95}$ reached 3×10^{-4}, more than $1{,}000\times$ below the fold-0 baseline, despite a clean loss decrease from 60 to 30. DETR-family decoders appear to require many more epochs to adapt to the elongated aspect-ratio geometry of RFI boxes.

Score calibration. Temperature-scaled post-hoc calibration applied to the final blend yielded at most $+0.0002$ mAP, consistent with the teacher ensemble already encoding well-calibrated confidence scores.

Inference-only SAHI. Sliced-inference SAHI [20] at 320-pixel tiles regressed mAP by -0.085, consistent with tile-boundary fragmentation of long horizontal stripes.

5. DISCUSSION AND CONCLUSION

The honest per-fold baseline saturates around 0.394 $\mathrm{mAP}_{50\text{-}95}$ (Table 1, top row): 2,753 images are too few to support reliable localisation of thin, low-contrast RFI stripes across the full IoU=0.50–0.95 sweep. Three findings drove the eventual lift to 0.4776. *(i)* Pseudo-distillation from a multi-source ensemble teacher contributed the largest single gain ($+0.041$): the soft signal in the teacher's confidence scores compensates for the small training set more effectively than the architectural alternatives we evaluated. *(ii)* Multi-resolution TTA between 608 and 704 px added a further $+0.034$ because thin stripes a few pixels tall are resolved unevenly by YOLO's strided feature pyramid at any single input size. *(iii)* Dropping the domain-overlapping fold-0 from the cross-fold ensemble — counter-intuitively, a *smaller* ensemble — was net-positive, because correlated predictions drag the WBF consensus when one source has seen the validation data during training.

Several axes that look attractive on leaked cross-fold validation did not transfer to held-out test: a second architectural family (RF-DETR) plateaued at $+0.0002$ across six checkpoint–weight variants; inference-only SAHI fragmented long horizontal stripes at tile boundaries (-0.085); and temperature-scaled calibration could not improve a teacher whose confidences were already well-conditioned by WBF fusion. Horizontal-flip TTA cost -6.5% relative, consistent with SAR side-looking geometry being asymmetric in azimuth–range [21]. The largest remaining gap is at high IoU thresholds, where sub-pixel height accuracy dominates and a 640-px input under-resolves the median $\approx$10 px stripe; ranked by expected return, the most promising next moves are (a) sliced *training* at 320-px tiles so the backbone sees RFI at doubled relative height, (b) a DETR-family detector with a different decoder trained for substantially more epochs than the ten that proved insufficient for D-FINE-X [22], and (c) a second pseudo-label refresh with V17 itself as the new teacher.

The final pipeline — YOLO26x with pixel-space NWD regression loss, ensemble-teacher pseudo-distillation, multi-resolution TTA and a selective cross-fold WBF — reaches 0.4776 $\mathrm{mAP}_{50\text{-}95}$ on the held-out test set; on meta-validation the ablation chain in Table 1 totals $+0.077$ over the per-fold baseline. The submission placed the *UdeM AI* team 28th of 120 teams on the public leaderboard of ESA ClearSAR Track 1. All training and fine-tuning were performed on a single cloud NVIDIA A100 GPU.

6. REFERENCES

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